Free pythagorean theorem worksheet worksheet 15

Maths | 8th grade | Mixed Operations

1. If x/3 - 2 = 7, what is the value of x?

A) 12
B) 21
C) 27
D) 30

2. What is the result of (5^2 - 4^2)/3?

A) 5
B) 7
C) 9
D) 15

3. If 4x + 5 = 3x + 11, find x.

A) 6
B) 12
C) 5
D) 7

4. Simplify the expression: 3(2x - 5) - 4(x - 3).

A) 2x - 3
B) 2x + 17
C) 2x - 7
D) 6x - 17

5. A rectangle has a length of 3x and a width of 2x. If the perimeter is 60, find x.

A) 5
B) 6
C) 7
D) 8

6. Solve for y: 2(y - 3) + 4 = y + 8.

A) 2
B) 3
C) 4
D) 5

7. What is 60% of 150?

A) 60
B) 75
C) 90
D) 120

8. If 2x - 3y = 12 and y = 2, find x.

A) 7
B) 8

C) 10
D) 12

9. Evaluate: (-3)^2 - 4 * (-2) + 5.

A) 21
B) 19
C) 23
D) 25

10. If the ratio of a to b is 3:5 and b = 20, what is a?

A) 8
B) 12
C) 16
D) 24

Answer Key

1. C
Add 2 to both sides to get x/3 = 9, then multiply by 3 to solve for x: x = 27.

2. A
Calculate 5^2 = 25 and 4^2 = 16; then subtract to get 9. Divide by 3 to get 3.

3. D
Subtract 3x from both sides to get x + 5 = 11. Subtract 5 from both sides to find x = 6.

4. A
Expand to get 6x - 15 - 4x + 12. Combine like terms to obtain 2x - 3.

5. A
Perimeter is 2(3x + 2x) = 10x. Set 10x = 60 and solve to get x = 6.

6. B
Expand to get 2y - 6 + 4 = y + 8. Simplify to get y - 2 = 8, thus y = 10.

7. C
Multiply 150 by 0.6 to find 60% of 150, which is 90.

8. B
Substitute y = 2 into the equation: 2x - 3(2) = 12. Simplify to 2x = 18, so x = 9.

9. C
Calculate (-3)^2 = 9 and 4*(-2) = -8. Then 9 + 8 + 5 equals 23.

10. D
Set up the proportion 3/5 = a/20. Cross-multiply to get 5a = 60, thus a = 12.